方鍾豪 Mike Fang

Lead Software Engineer / Team Leader

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SUMMARY

Engineering leader with 10+ years across cybersecurity, media tech, ad tech and HR tech. I lead full-stack teams (TypeScript, Next.js, NestJS, PostgreSQL, AWS), design scalable systems, and grow engineers through hiring, coaching and clear technical direction. Today I am the sole lead engineer of an AI-powered recruiting CRM, and I run an AI-first engineering workflow with Claude Code and Cursor. Open to conversations about engineering leadership and software product engineering.

EXPERIENCE

Lead Software Engineer — Bruce HR Consulting

Taipei, Taiwan · 12/2024 – present

I built company's internal CRM alone, from scratch, and I still run it: backend, frontend, database, AI, cloud and deployment. Used daily by consultants, sales and leadership across tens of thousands of candidate profiles.

Product impact

- Built the **recruiting Kanban** that consultants work from every day, tracking each candidate–job pair from submission to onboarding.
- Built a **leadership command board** for ranking clients and openings by priority and tracking each consultant's weekly progress.
- Built this CRM **alone, from scratch**, and still run it. It's used daily by consultants, sales and leadership, manages a candidate database of **tens of thousands of profiles**, and covers the whole hiring flow from sourcing to placement.
- Built a **Salary Report** that shows pay against years of experience across the candidate database. Consultants can filter by job function, location, project, job opening and date, remove outliers, and export the chart. They use it to give clients and candidates salary advice backed by data.
- Replaced spreadsheet tracking with a **Sales Sheet** module that keeps consultant targets, deals and replacement hires in one place.
- Built **pipeline metrics** that help the team **forecast sales numbers**.
- Built market-research tools that refresh on a schedule: an **AI reader for earnings-call transcripts**, an **industry intelligence board** and a **tracker for US job-posting trends**.

AI & search features

- Built **AI candidate–job matching** that finds the **top 100 candidates for a job opening across the entire candidate database** in one click. It compares candidates and the job description on 5 areas (must-have skills, domain experience, functional experience, education, overall profile), combined with hard requirement filters and normalized skill names.
- Kept matching **fast and cheap**: similarity is calculated inside the database (pgvector), and each candidate's text embeddings are reused until their profile changes. Running a match again costs **almost nothing in AI spend**.
- Built an **AI resume-vs-job match review** that gives a match score with reasons for and against, strengths, red flags and suggested improvements. Consultants can then ask the AI follow-up questions about the match.
- Launched an **AI Data Explorer** that lets non-technical staff ask questions about CRM data in plain English, without filing report requests.
- Built an **AI Resume Builder** that saves consultants about **40% of the time** they spend writing and managing resumes.
- Built **AI usage and cost dashboards** with per-user breakdowns, so leadership can see and control LLM spending for the first time.
- Reduced unnecessary AI token spend by **10%** by choosing a model that fits each feature.

Engineering & performance

- Made backend builds **6× faster (SWC)**, linting **12× faster (oxlint)** and frontend builds **about 23% faster**.
- Sped up page loads across the main dashboards by running server requests in parallel, caching shared reference data and loading heavy tables and charts only when needed, guided by Lighthouse audits.
- Every feature starts from a written spec, and changes must pass **435 unit-test files**, Playwright E2E tests and GitLab CI.

Infrastructure & security

- Brought all production AWS resources under **Terraform** by myself, so the whole setup is version-controlled and can be rebuilt from code instead of being clicked together by hand.
- Set up automated weekly backups stored in a second region and reliable monitoring alerts.
- Tightened security across the stack: limited each service's access to only what it needs, locked down network access rules, and hardened server settings against common attack paths.

AI-assisted engineering (Claude Code & Cursor)

- Run an AI-first workflow where Claude Code and Cursor follow **one shared instruction file** and a shared skill library (**20 Claude Code skills and Cursor rules**). The skills cover spec-first design, safe database migrations, security and performance audits, and verification before any change counts as done. Agents never see secrets and get least-privilege access to data.

Senior Full Stack Engineer — Placements.io

Taipei, Taiwan · 11/2022 – 10/2024

Advertise Order Management System

- Develop feature with React/Redux with TypeScript as frontend and Ruby on Rails as backend, collaborating with U.S. team
- Supported a new AD server integration end to end.
- Made database queries faster by tuning indexes.
- Made frontend loading faster by reducing the bundle size with lazy loading.

Lead Software Engineer — VYOND

Taipei, Taiwan · 01/2022 – 10/2022

Online SaaS Animation Software Tool for Businesses

- Being a project leader to develop new features of our SaaS product with engineers, designers, data specialists, animators and PMs between Taiwan and Hong Kong
- Hire talented engineers into teams and provide coach guideline
- Review the technical specification and provides suggestions to engineers
- Collaborate with machine learning team to provide AI features to users.

Senior Software Engineer — VYOND

Taipei, Taiwan · 01/2020 – 12/2021

Online SaaS Animation Software Tool for Businesses

- Being a project leader to develop new features of our SaaS product with engineers, designers, data specialists, animators and PMs between Taiwan and Hong Kong
- Improve product system stability and performance
- Cultivate new comers to make a good working environment in Taiwan Office
- Design and implement a request scheduler system to handle burst video generation requests request and reduce 60% unnecessary requests
- Lead a new version of video management project to provide better user experience. Including code refactor and unit tests
- Head a content share mechanism project to solve customer pain points

Software Engineer — VYOND

Taipei, Taiwan · 09/2018 – 12/2019

Online SaaS Animation Software Tool for Businesses

- Develop new features with engineers, designers, data specialists, animators between Taiwan and Hong Kong
- Lead a customer issues task force and propose solution to enhance our product
- Reduce 80% unnecessary search results in the search improvement task force
- Increase our retention rate about 40% in the onboarding improvement task force

Senior Software Automation Engineer — NETGEAR

Taipei, Taiwan · 05/2018 – 07/2018

Smart Home Solution

- Maintain and improve existing REST API automation framework. Save 20% time of daily regression
- Payment flow API integration testing and daily regression trouble shooting
- Load/Volume testing for the web server
- Cooperate with colleagues from USA/India/Russia/Taiwan

Senior Software Engineer — Trend Micro

Taipei, Taiwan · 01/2016 – 05/2018

Dynamic Sandbox Analysis for Unknown Malware Detection

- Design and develop the process of sandbox module update to increase deployment flexibility and decrease the response time for detecting a new threat
- Investigate and adopt multiple hypervisors and networking solutions to sandbox environment to improve overall performance
- Design and develop a framework for sandbox result analysis and integrate core techniques modules to improve malware detection rate
- Maintain a Windows application to help customer creates their customized virtual environment
- Maintain CI Jenkins system and write unit test
- Cooperate with the Technical Support team to solve customer feedback issues.
- Act as a communication bridge between developers and QA engineers

Software Engineer in Test — Trend Micro

Taipei, Taiwan · 09/2013 – 12/2015

Dynamic Sandbox Analysis for Unknown Malware Detection

- Have code review/end-to-end testing to find potential defects and drive the architecture design
- Make test malware samples using C/C++, C#...etc. to trigger specific behavior for integration testing
- Build an automated fast build verification testing (BVT) system as a part of CI system to verify basic build quality. Remove unnecessary manual basic checking
- Create a web control system using Python Flask framework to integrate existing testing systems
- Re-design and enhance existing automation framework. Remove limitations of old framework and fulfill requirements from team members and increase test automation rate by at least 20 %
- Coach new comer and lead them to be familiar with our project and culture
- Get top score in 1-month virus training in Trend Micro
- Act as a proactive contact window between Taiwan team and China team

Part-time Programmer — CAMEO Info Tech

HsinChu, Taiwan · 07/2011 – 04/2013

Android and iOS Application Development

- Interactive Picture Book with Corona SDK in Lua language
- An native iOS APP for WIFI storage with WebDAV protocol with Objective-C
- A native Android hot-spot setting widget via NFC

EDUCATION

Master of Engineering / Computer Science — National Chiao Tung University

Hsinchu, Taiwan · 09/2011 – 07/2013 · GPA 4.3/4.3

Thesis: A Decision-theoretic Data Prefetch Algorithm for Reducing Latency and Amortizing Traffic of Mobile Applications.

Bachelor of Engineering / Computer Science — National Chiao Tung University

Hsinchu, Taiwan · 09/2007 – 06/2011 · GPA 4.16/4.3

SKILLS

General: Web Application Development, System Software Development, Test Plan and Cases Design, Automation Testing, Regression Testing, Performance Testing Design and Improvement, Continuous Integration and Delivery, Agile and Scrum experience, Certified Scrum Master, Trend Micro Certified Cyber Security Professional, Members mentoring, Communication, Project Planning, Managing and Leading, Full-stack Development, System Design, Access Control, Mobile Application Development, Quick Learning

Programming Languages: JavaScript, Python, Shell Script, Java, C/C++, C#, Perl, Ruby, TypeScript, Lua, Objective-C

Web Developments: React.js, Redux, Node.js, NestJS, Express.js, Flask, MySQL, Redis, HTML5, CSS3, SCSS, Gatsby.js, Next.js, Astro, Restful API, Open API, AVA, Cypress, AWS Services, ELK, Ruby on Rails

AI: LLM Application Development, Text Embeddings, Semantic Search, pgvector, Model Evaluation, Claude Code, Cursor

Tools & Platforms: Linux, Windows, MacOS, git, Perforce, VSCode, Lighthouse, QEMU, VirtualBox, Docker, Jenkins, Postman, Jmeter, Vyond, Trello, Jira, Selenium, PostgreSQL

Natural Languages: English (Business Fluency, TOEIC 940), Chinese (Native Speaker), Japanese (Beginner, N4)